Game design document

Rewritten 2026-10-28 for the game as it plays today in `ArknightsTest`. The first version (2026-10-12) described a 24 × 24 board with a random tower shop; since then the game has been reworked in rounds after play tests (cards, tower levels and gear; a generated map with several entries; towers on walls; the ore economy and the build menu; power; upgrade paths; wall tiers and breakable mountains; a development director). Each round is an entry in `DEVLOG.md`, and the first version is in git (`132d8ca`). No name has been chosen for the game yet (§12, question 12).

This file describes the game from the player's side: what is on the screen, what the player does, and what the finished game is meant to be. The other documents answer other questions: `CLAUDE.md` says what the thesis claims and what must never be broken, `ARCHITECTURE.md` how the code is built, `WORKPLAN.md` in what order, and `Docs/proposal/` is the thesis proposal. Where this file and `CLAUDE.md` disagree, `CLAUDE.md` wins.

Every part of this file carries one of four marks:

Mark	Meaning
BUILT	In the game today, and tested.
PLANNED	Decided and in the work plan. Not built.
PROPOSED	A suggestion made in this file, so that there is something concrete to look at. Not decided. Accept it, change it or drop it.
OPEN	Nobody has decided. §12 lists every open question, with who must answer it and by when.

Every number in this file is a first guess checked by the next tester session, not a balanced value. The game's numbers live in `Assets/Scripts/Config/SimConfig.asset` (exported to `Maps/SimConfig.json`) and the towers in `Maps/Towers.json`.

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1. The game in short

A single-player maze tower defense on a map generated from a seed. You place your core, claim ore deposits with collectors, draw walls to bend the enemies' route, and put towers on your walls and on the rocks to shoot them. Enemies come in from the edges of the map, one to three entry points a wave, and head for your core, or for your collectors.

The unusual rule: **nothing you build blocks the path.** Every wall, tower, collector and even the mountains can be chewed through. An enemy takes the cheapest way to its goal, and "chew through this" is priced into that choice like any detour. So you never close a route; you make it long and expensive, and you put towers along it.

Waves come one after another, each stronger than the last. Before each wave you build, upgrade, move and sell; the wave starts when you press Start. The session ends when the core has lost its ten hit points.

What the thesis adds. The game exists in two versions that differ in one thing only. In the *static* version the waves follow a fixed plan by wave number. In the *adaptive* version a wave director looks at what you built, picks the kind of wave that presses on that build's weak point, and tells you so before the wave. Both versions send waves of the same total strength. The claim to be tested is that the adaptive version keeps a player rebuilding, where the static one lets them find one good layout and repeat it.

Design pillars

1. **You bend the path, you never close it.** A path to the core always exists.
2. **Three levers, one resource.** Towers are damage. Walls are time under fire. Collectors are income, and they stand where the ore is, away from the core. Walls alone kill nothing.
3. **Every wave asks a question about your build**, and says the question out loud.
4. **Fair.** Whatever the director sends has the same strength as the static wave of that number, and the director never touches your prices, your towers or your ore (§7.4).
5. **Readable.** The routes are drawn (§6.4), the next wave is shown before you start it, a tower shows the tiles it reaches before you pay, and an enemy shows its health.

2. One session, one wave

```
BEFORE WAVE 1    place the core anywhere inside the gold frame (Done: the core stays here)
  ▼
BUILD PHASE      no clock: it lasts until you press Start
  │              the next wave is already planned and shown: its enemies, its route lines,
  │              and (adaptive version) the director's line in the banner
  │              build · upgrade · move towers · sell · research · undo and redo
  ▼
THE WAVE         enemies come in at one to three edge points
  │              towers fire · enemies walk the route, or chew through what is in the way
  │              you may still build; you may not sell or move towers
  ▼
THE WAVE ENDS    when the last enemy has been killed or has reached its goal
                 income is paid · the next wave is planned · a new build phase begins
```

BUILT, all of it. A wave can be restarted from its start at any time after it began, with everything you built for it still standing (RESTART WAVE n, or PLAY WAVE n AGAIN in the build phase after it).

3. The map

3.1 Generated from the seed

BUILT (2026-10-13, enlarged 2026-10-20). Every session starts from a seed, and the map is made from it (`MapGenerator`):

- **48 × 48 tiles.** A two-tile ring at the edge is the enemies' ground: nothing is built on it.
- **22 ridges of rock** (mountains, §3.3), so the board is never empty and never the same.
- **28 ore deposits** in clusters of four, at least one cluster near the middle.
- **Entry points** all round the edge. A wave uses one in the first waves, two from wave 4 and three from wave 8, each with a walk of at least 12 tiles to the core and at least 10 tiles apart.

Today every session gets a new random seed. PLANNED the study uses two fixed seed sets, crossed with the condition and the order (decision D9), so both versions are played on the same maps.

3.2 The view

BUILT An orthographic camera tilted 52° from the horizon, so towers read as figures standing on the board; the board keeps one tile size from the near edge to the far one. The mouse wheel zooms toward the mouse, WASD or the arrow keys move the view.

3.3 Mountains

BUILT (breakable since 2026-10-28). The rocks are mountains: very tough terrain.

- A tower may stand on a mountain (§5.2). Nothing else is built on one.
- To an enemy on foot a mountain counts like 60 tiles of walking and takes 60 seconds to chew, ten times a plain wall. It goes round unless going round is longer still. A sapper pays the full price too, so sappers do not tunnel the same mountain every time.
- A broken mountain is gone for the rest of the session. It cannot be sold and pays nothing. A tower standing on it is destroyed when it is chewed through; the mountain under it stays.
- Flyers pass over mountains.

4. Ore

BUILT (the ore economy, 2026-10-20). There is one resource, ore. You start with 80.

	Ore
At the start	80
After every wave, from the core	+4
After every wave, from each standing collector	+2 near the middle, up to +4 near the edge
A collector	−6
Kills, chewed walls, destroyed towers	nothing
Selling something bought in this build phase	all of it back
Selling something older	half back
A collector destroyed by enemies	half its price back at once
Repairing the core	−10 for 3 hit points

- **A deposit holds 30 ore**, then it runs dry. A collector on an empty deposit earns nothing, and an empty deposit takes no new collector. The deposit shows what is left.
- **A collector is terrain**, like a wall: it counts like 20 tiles of walking and takes 8 seconds to chew. A destroyed collector pays nothing for that wave.
- **Power.** A tower fires only while it is powered. The core powers up to 4 towers within 4 tiles, each collector up to 2 within 3, a pylon up to 2 within 3 (§5.5). As many towers as possible get power. Selecting a source shows the tiles it reaches and lines to the towers it powers; clicking a tower in its reach ties the tower to it. When raiders destroy a collector, the towers it powered go dark at once.

What it is for: the stack of towers round the core, which beat everything in the first play tests, now has a cost. Income stands where the ore is, spread over the map, and has to be defended or lost.

5. What you build

Everything is bought from the **build menu** in the bottom bar, by category: TOWERS, ECONOMY (collector, pylon, research, repair), GEAR and ORDERS (walls, sell, move, upgrade, target). Hovering a button shows its card. **There is no random shop** since 2026-10-20: every price is fixed and everything is on offer from the start, except what research unlocks.

5.1 Walls

BUILT A wall is one tile, drawn with a click, or a line of them with a drag (Q).

Wall	Price	To an enemy on foot it counts like	Seconds to chew
Wall	1	15 tiles of walking	6
Stone wall	3	30	18
Fort wall	5	45	36

A wall goes on any free tile that is not a mountain, a deposit, the core or the edge ring. The tile under the mouse shows whether the spot is legal before you click; the preview and the click use one rule, so they can never disagree. A sapper pays a fifth of a wall's price (a plain wall counts like 3 tiles to it).

- **Thickness is the first thing to watch.** One plain wall tile costs a walker 15 tiles' worth to cross; two cost 30.
- **Length is the second.** Walkers go round only while going round is cheaper than chewing. Stretch the detour too far and they chew through the thinnest spot instead.

5.2 Towers

BUILT A tower stands **only on a wall or a mountain**: build the wall first, then the tower on it. It fires only while powered (§4). It picks its target by its target order (first, strongest, weakest, nearest, raiders first; T), hits at once, and waits for its next shot.

Tower	Price	Damage type	Damage	Every	Range	Also	Hits flyers
Archer	10	physical	6	0.5 s	4		yes
Ballista	14	physical	32	2 s	5		no
Cannon	12	fire	6	1.2 s	3	splash 0.75	no
Brazier	10	fire	0.8	0.1 s	2	splash 0.5	no
Frost	10	frost	0.5	0.6 s	2.5	splash 1; slows to 40 % for 1.5 s	no
Sniper	18	physical	40	3.2 s	8		no
Shard	14	frost	5	0.6 s	3.5	three targets	no
Flak	14	fire	2.5	0.8 s	4.5	splash 0.75	yes

Every tower can be built from the start. Towers can be **moved** freely in a build phase (M), and **sold** (X).

5.3 Upgrades and research

BUILT (2026-10-27). Every tower has **three upgrade paths of two steps**. The first upgrade picks a path; the second can only be that path's next step. A tower with one upgrade is rare, with two epic. The first upgrade costs 12 ore, the second 20.

Research opens upgrades: at tier 1 no tower can be upgraded, tier 2 (15 ore) allows a tower's first upgrade, tier 3 (30 ore) its second.

Tower	Paths (first step / second step)
Archer	Ranger: shoots 2 / 3 enemies · Hawkeye: ×2 / ×3 damage to flyers · Marksman: every 3rd shot ×3 / ×5
Ballista	Breaker: ignores armour / pierces 3 · Executioner: finishes under 20 % / 35 % · Skyward: hits flyers / fires 40 % faster
Cannon	Mortar: +2 range / bigger blast · Incendiary: sets burning · Concussion: slows
Brazier	Inferno: more damage, burning · Bonfire: pulls enemies within 4 tiles (lure) · Smoke: slows, then exposes
Frost	Deep freeze: slows to 25 % · Brittle: enemies take +25 % / +50 % · Ice spikes: flat damage
Sniper	Assassin: every 2nd shot ×2, execute · Anti-air: hits flyers · Spotter: ignores armour, exposes
Shard	Splinter: more shards · Frostbite: slows, then brittle · Lance: one big shard
Flak	Barrage: faster, two targets · Skybreaker: ×2 / ×4 damage to flyers · Fire shells: burning, heavier
Pylon	Grid: powers 1 / 1 more · Relay: reaches 1 / 1 tile further · Beacon: collectors nearby pay +1 ore

5.4 Gear

BUILT (flat gear since 2026-10-27). A tower has gear slots; gear is crafted from the GEAR tab at half its list price and dragged onto a tower.

Gear	Does	Ore
Scope	+1 range	4
Far scope	+2 range	9
Edge	+1 damage a second	4
Keen edge	+3 damage a second	10
Extractor	takes a tower's gear off	2

Damage gear adds damage *a second*, once a shot to the first target and never multiplied by a crit or a volley, so it is worth the same on every tower.

5.5 The pylon

BUILT 18 ore. It does not shoot: it powers up to 2 towers within 3 tiles, and has power itself. It may not stand within 5 tiles of the core (cheap power for outposts, not for a stack round the core). Its upgrades add one tower or one tile each (at most 4 towers, as the core).

5.6 Changing your mind

BUILT Undo and redo (Ctrl+Z, Ctrl+Y) go back up to 30 steps within the build phase. Anything sold in the build phase it was bought in gives all its ore back.

6. Enemies

6.1 What an enemy is

BUILT An enemy has hit points, a walking speed, a digging speed, and one resistance per damage type. It ends in one of two ways: **killed** (nothing is paid) or **through** to the core (the core loses 1).

6.2 Three ways of moving

BUILT (WP-C2). This is what the director's wave types are made of.

Class	How it gets to its goal	Digs	Which towers can hit it
Walker (ground)	The cheapest way, counting a plain wall as 15 tiles of walking. Usually goes round.	Only when going round costs more.	All
Sapper	The same, but your walls count a fifth (mountains full price). Goes through walls that walkers go round.	Often, and faster.	All
Flyer	A straight line to the core, over everything.	Never.	Only towers that hit flyers

Two behaviours come on top of a class:

- **Raiders** go for your collectors, the nearest first, then the next, then the core.
- **Siege enemies** stop as soon as a tower is within 4.5 tiles, set up for 2 seconds, then every 1.5 seconds shell the tower with the most towers round it, and the towers next to it. A tower that has taken 10 is knocked out until the next build phase, never destroyed. Towers that outrange them (ballista, sniper, range upgrades), slows and spreading out are the answers.

A tower with the lure ability (the brazier's Bonfire path) pulls enemies on foot within 4 tiles.

6.3 Enemy types

BUILT Each type is a set of numbers on top of the plain walker of that wave.

Type	Class	Compared with the plain walker
Basic	walker	the baseline
Swarm	walker	twice as many, 0.4 the hit points, 1.3 times as fast, closer together
Runner	walker	0.8 as many, 0.6 the hit points, 1.7 times as fast
Brute	walker	a quarter as many, 4 times the hit points, 0.7 as fast
Armored	walker	takes 35 % of physical damage; 1.2 times the hit points
Ember	walker	takes 25 % of fire damage
Sapper	sapper	half as many, 1.5 times the hit points, digs 4 times as fast
Flyer	flyer	half as many, 0.35 the hit points
Bats	flyer	many, 0.15 the hit points, 1.2 times as fast
Raider	walker, raids	half as many, 1.2 times as fast, digs 1.5 times as fast
Siege	walker, siege	6 times the hit points (3 times at first, §7.2), 0.6 as fast
Frostborn	walker	immune to frost; kept for the tests, no longer sent (too strong)

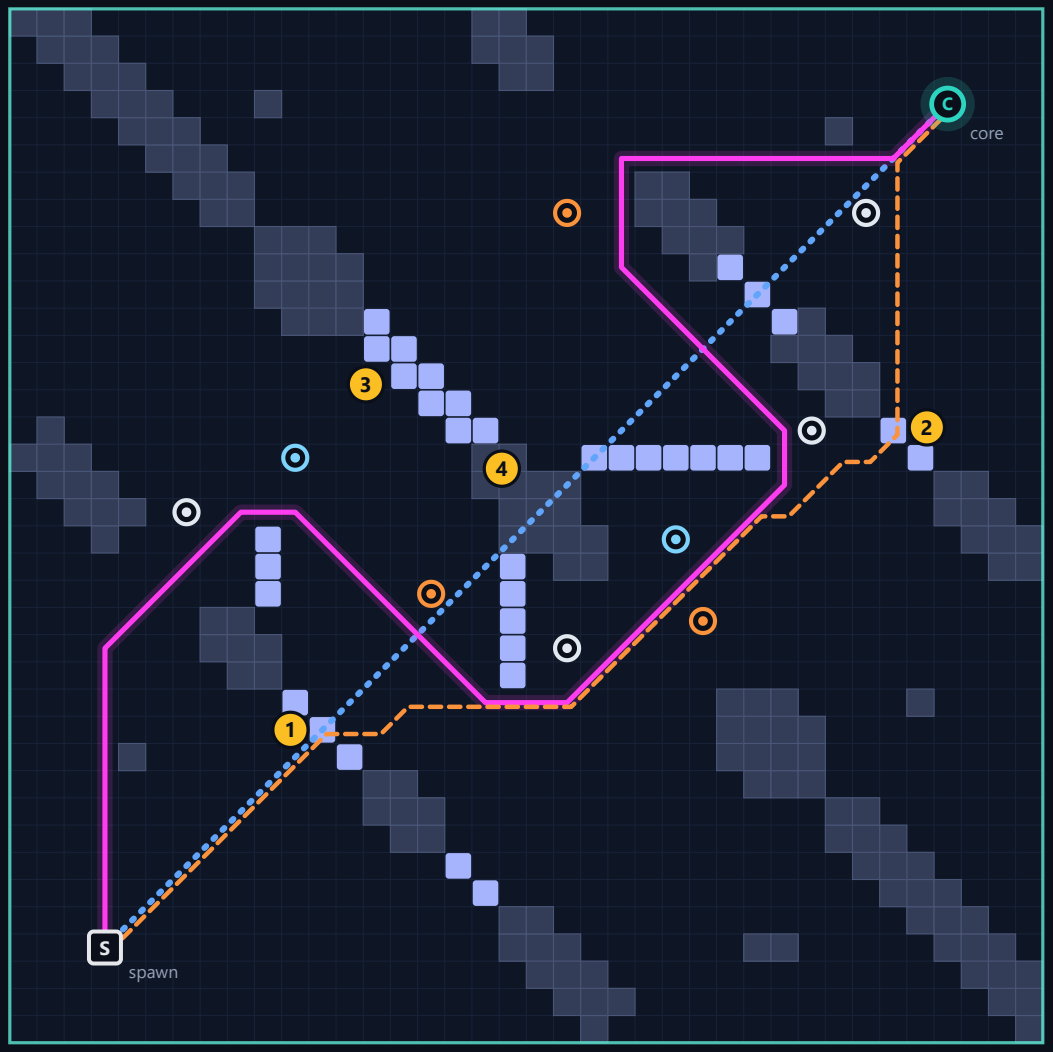
6.4 Where enemies walk FIGURE 1

BUILT The picture is an example board from the first version of this file; the rule it shows is unchanged. The pink and the orange route were not drawn by hand: they are what the game's own path rule gives for that board.

Nobody assigns a path to an enemy. There is one rule, and every enemy follows it on its own: *at every step, go to the neighbouring tile from which the rest of the way to my goal is cheapest for my kind.* A flyer does not use the rule at all; it flies a straight line.

- **Same kind, same entrance, same route.** The walkers of one entry walk one line.
- **Different kinds, different routes, on the same board.** In the picture the walkers go round every wall (about 71 tiles, past 9 towers), the sappers through the two thin walls (about 50 tiles, past 5), the flyers straight over (44 tiles, past 1). One layout is several different problems, depending on what arrives. That is what the director works with.
- **The director chooses the kind and the entries, never the path.** Wherever an enemy goes, you could have read it off your own board.
- **A route can change in the middle of a wave**, when something is chewed through or you build.
- **A route is never closed.** Wall off every gap, and the wave chews through the cheapest spot.

FIGURE 1 **6.4 Where enemies walk**



Where enemies walk

One board, three kinds of enemy, three routes. Nobody steers an enemy: each one takes the cheapest way to the core for its own kind.

Walkers go round

To a walker, a wall tile counts like 15 tiles of walking. Going round is cheaper here, so they take the long way: about 71 tiles, past all nine towers. They chew nothing.

Sappers go through

To a sapper, a wall tile counts like 3 tiles. It cuts through the two walls that are one tile thick (1 and 2), leaves the thick wall (3) alone, and is there after about 50 tiles, past five towers.

Flyers go straight

A straight line over rocks (4), walls and towers: 44 tiles. Nothing you build bends it. Only towers that can hit flyers shoot at them; here one of those is in reach.

Same kind, same entrance, same route.

Enemies of one kind walk in single file along one line. The line is drawn on the board before the wave, so you can build against it.

The director chooses the kind, never the path.

It decides what comes (and later, from which side). After that every enemy finds its own way by the rule above. It cannot aim one at a tile.

A route can change in mid-wave.

When a wall is chewed through, or you build, every enemy on the board works out its way again from where it stands.

Dark blocks: rocks, part of the map. Light tiles: walls the player drew. Rings: towers. The pink and orange routes are what the game's own path rule gives for this board.

Where the three kinds of enemy walk on one board

What the screen shows. BUILT in a build phase, one line for each kind of enemy in the next wave, from its entry; raiders' lines go to the collector they reach first. During a wave, the lines of the kinds in that wave. The lines follow the enemies' own step rule, so they bend to a lure as the enemies do.

7. Waves

7.1 The static version

BUILT as the baseline the director starts from. The plain walker of wave *n*: 12 enemies in wave 1 and one more every two waves; hit points 30 + 4.5 a wave, then 4 % more a wave on top; speed 1.25 tiles a second, 0.0375 more a wave, at most 2.25.

Each wave is made of one to three types in a fixed twelve-wave cycle (basic; runners; sappers; armored; flyers; swarm; brutes and basic; ember and sappers; bats; swarm and runners; armored and flyers; brutes, swarm and bats). On the ore map raiders come too, from wave 4, every few waves. OPEN **(question 21):** whether the study's static version keeps this cycle, or sends plain walkers only.

7.2 The development director

BUILT (2026-10-26, rewritten 2026-10-28). What the game runs today. It is **not** a study condition and **not** the thesis director: it has no learning, no build profile and no price list that keeps every wave type equally strong.

At the start of each build phase it reads the board and gives each kind of wave a score; the highest is sent, and a kind sent last wave loses a little:

Wave	What arrives	Sent when
Baseline	the wave of §7.1	nothing below scores higher
Outpost raid	raiders, 15 to 40 % of the wave	collectors that no tower covers (from wave 4)
Siege stack	one to three siege enemies	most of the damage comes from one cluster of towers (from wave 8; one alone for the first three siege waves)
Resist mix	armored or ember enemies, 40 %	one damage type is over 60 % of your damage
Air raid	flyers, 40 %	less than a quarter of your damage can hit flyers (never when none can)
Maze breach	sappers, 30 %	a long wall maze: 10 or more walls, and the route 1.4 times the straight line (the only time sappers come)

At least half of every wave still goes for the core. The banner shows one written line for the chosen kind, e.g. "Raiders are coming for your undefended outposts."

7.3 The adaptive version

PLANNED (WP8 to WP11).

What the director looks at. At the start of each build phase it measures five things:

Measure	In plain words
maze length	how much longer the route is than on an empty board
breach vulnerability	how much of the route depends on walls or towers that could be chewed through
chokepoint reliance	whether your damage is done in one spot or spread along the route
tower concentration	whether your towers stand together or apart
damage type mix	whether your damage is one type or several

Their exact formulas are OPEN (question 5).

What it can send. One of a small set of wave types, each meant to press on one weakness: the proposal's four (breach the thin wall, swarm the chokepoint, split groups, flying bypass) and the development director's kinds (§7.2) are the candidates. The final set is OPEN (question 4). Which builds each type really moves a player to change is not a rule of the game: it is what the study measures.

How it chooses. It starts from written scores, as the development director does. On top of them it learns, during your session, which wave types actually made *you* rebuild, and nudges close calls that way, by at most one point. It is rewarded for a rebuild, not for beating you: a wave that ends your session scores badly, because a player who is out builds nothing more.

7.4 What keeps it fair

Fixed rules of the project (`CLAUDE.md` §4):

- **Same strength.** Every wave type costs exactly what the static wave of that number costs, on one shared price list. The director chooses the shape of a wave, never its size.
- **Nothing that cannot be answered.** Flyers are not sent while no tower can hit them.
- **No repeats in disguise.** A wave too similar to the previous one is ruled out.
- **Hands off your side.** The director never changes a price, a tower or your ore. OPEN **(question 6):** raiders destroy collectors and siege enemies knock towers out, so a wave type that sends them changes your income and firepower, which the static version does not.

7.5 What is the same in both versions

The map and its seed, every price, the strength of every wave, and the screen. Only the shape of the waves differs, and with it the banner's line and which route lines are drawn.

8. Losing, restarting, a session

BUILT The core has 10 hit points and loses one for every enemy that reaches it. At zero the game is over: "You reached Wave N", and Restart. There is no way to win today.

BUILT a wave can be restarted any number of times (§2). OPEN **(question 22):** whether a participant may restart waves in the study, since a restart replays the director's choice and changes how many waves count.

OPEN **(question 1): how long is a session?** The study compares the waves a player completed in each of their two sessions, cut to the shorter one, so it needs a stated length (about 25 waves is assumed) and a game in which an ordinary player gets most of the way there.

PLANNED **(WP14): a study session.** The participant types a code on a start screen; the code decides their two sessions (one static, one adaptive, two different seeds, in a counterbalanced order).

9. What the screen looks like

9.1 Today

BUILT The pictures are from the game on 2026-10-28.

Placing the core FIGURE 2

Brown discs are ore deposits, dark tiles mountains; the pink line is the first wave's route.

A build phase FIGURE 3

Top left: the core's health and the wave. Top centre: the banner. Top right: ore and income, powered towers, research tier; under them the route key, the next wave's enemies as chips, and the damage panel. Bottom: the build menu by category, the gear slots, undo and redo; bottom right pause, speed (x1, x3, x6) and the wave button.

The director's warning FIGURE 4

Unguarded collectors drew a raid: the banner says so, the chips show 7 armored and 2 raiders, and the raiders' route (green) runs to the collectors.

A wave FIGURE 5

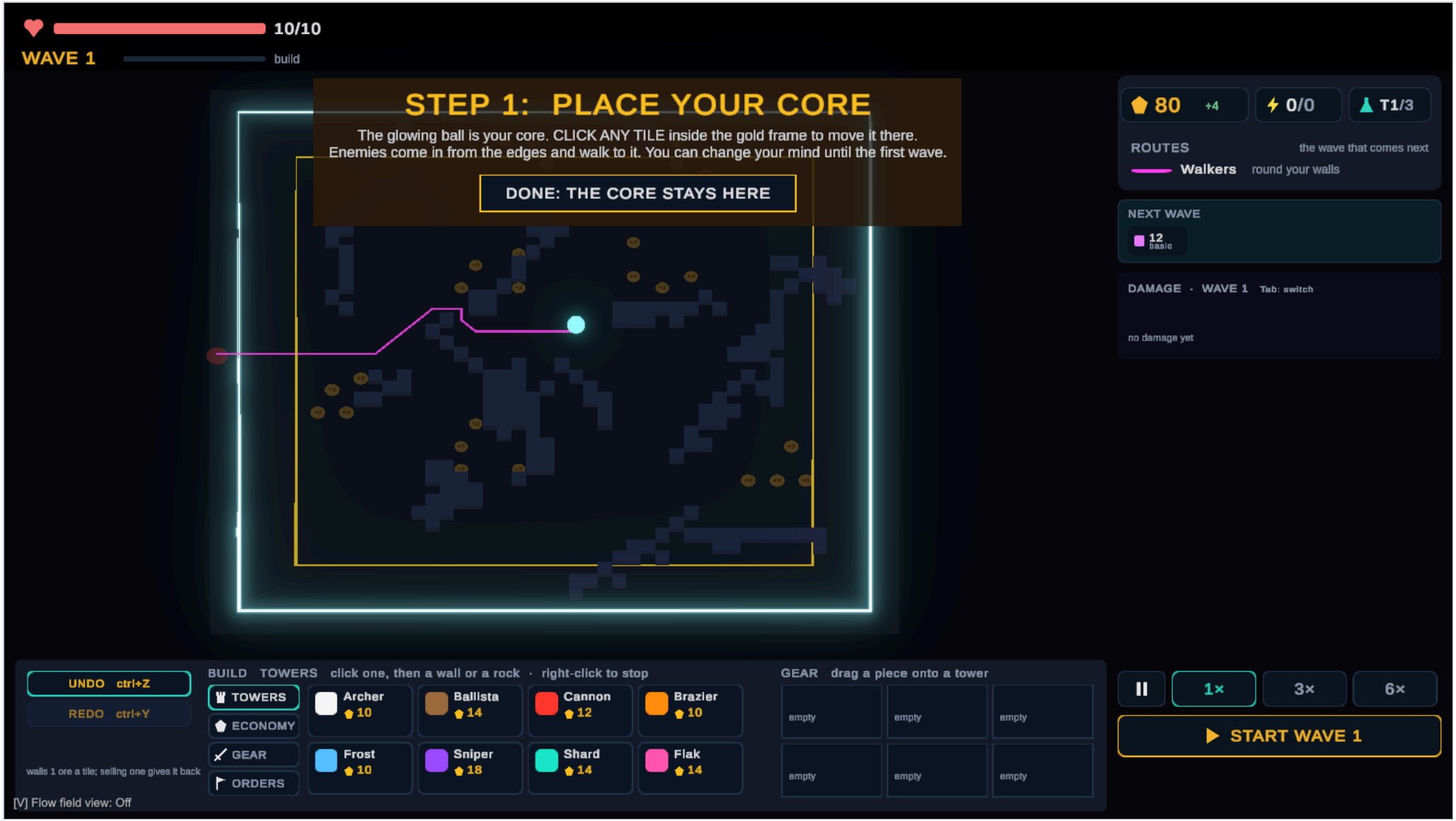
After a wave FIGURE 6

The damage panel shows each tower's share; the three best are named at the top for a few seconds when a wave ends.

The flow-field view (V) FIGURE 7

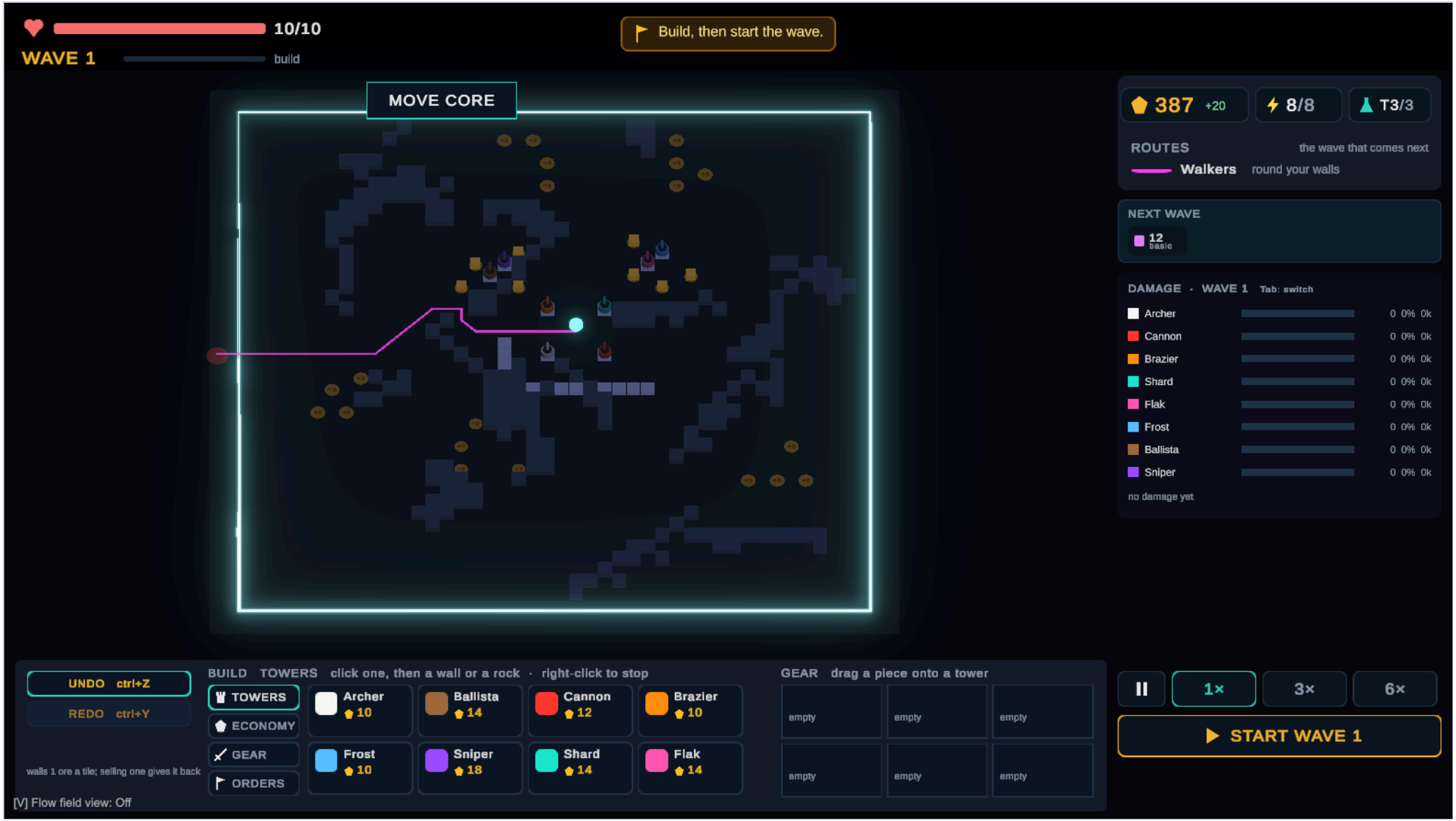
For development: the cost to the core of every tile, as colour, and the way out of it, as an arrow.

FIGURE 2 **Placing the core**



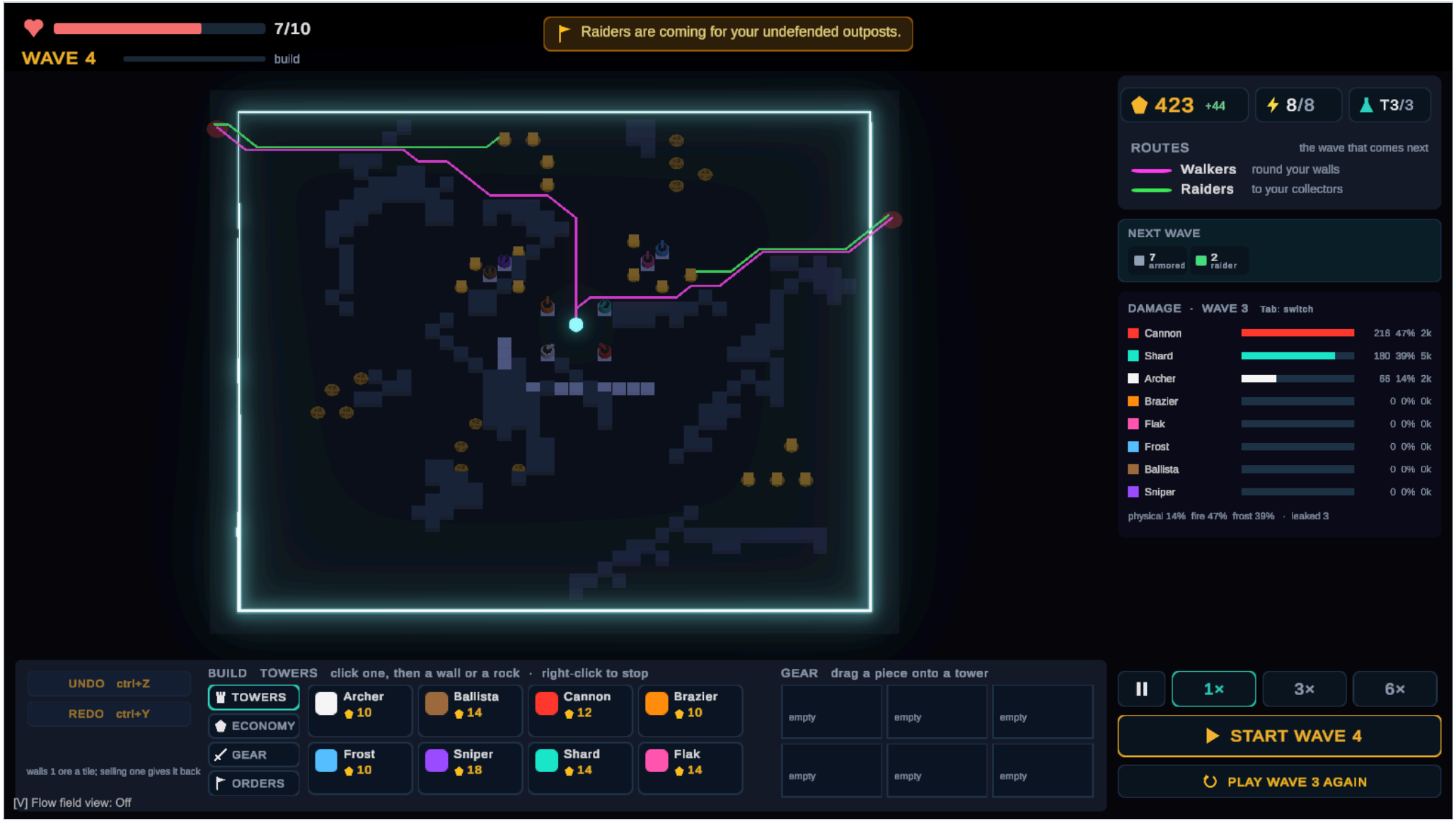
Session start: the player places the core inside the gold frame

FIGURE 3 **A build phase**



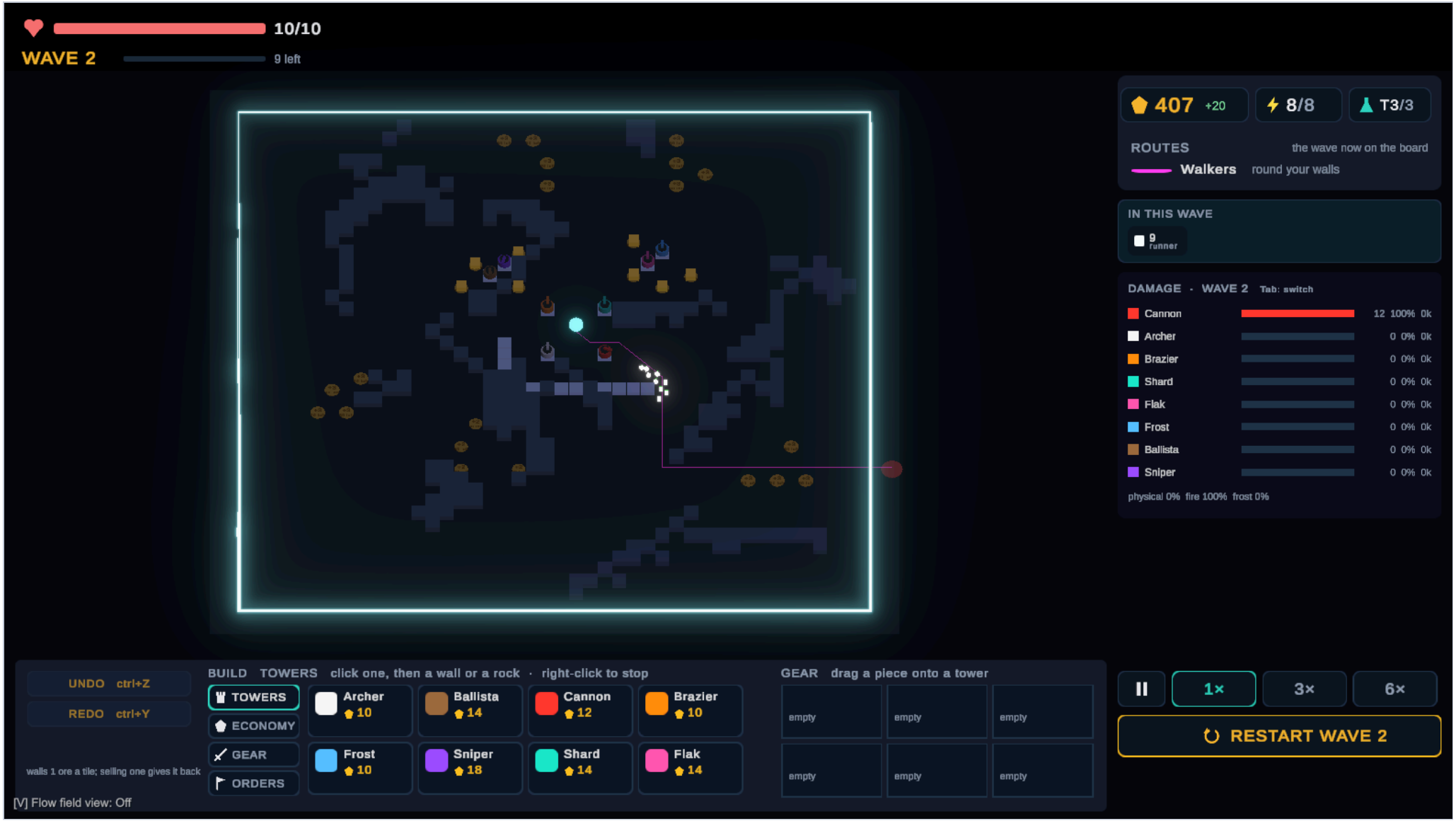
A build phase

FIGURE 4 **The director's warning**



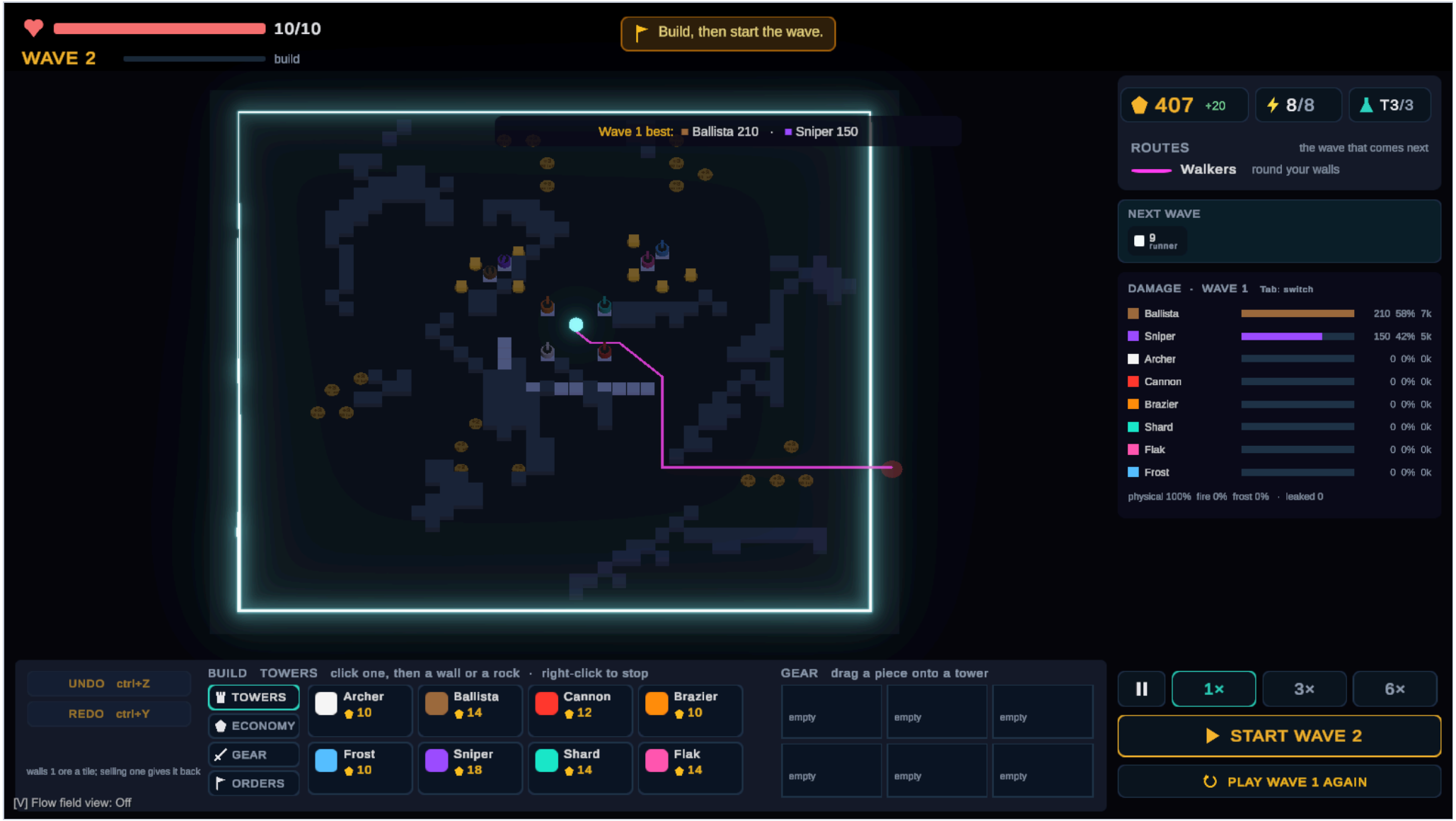
A planned raid: the banner, the enemy chips and two route lines

FIGURE 5 **A wave**



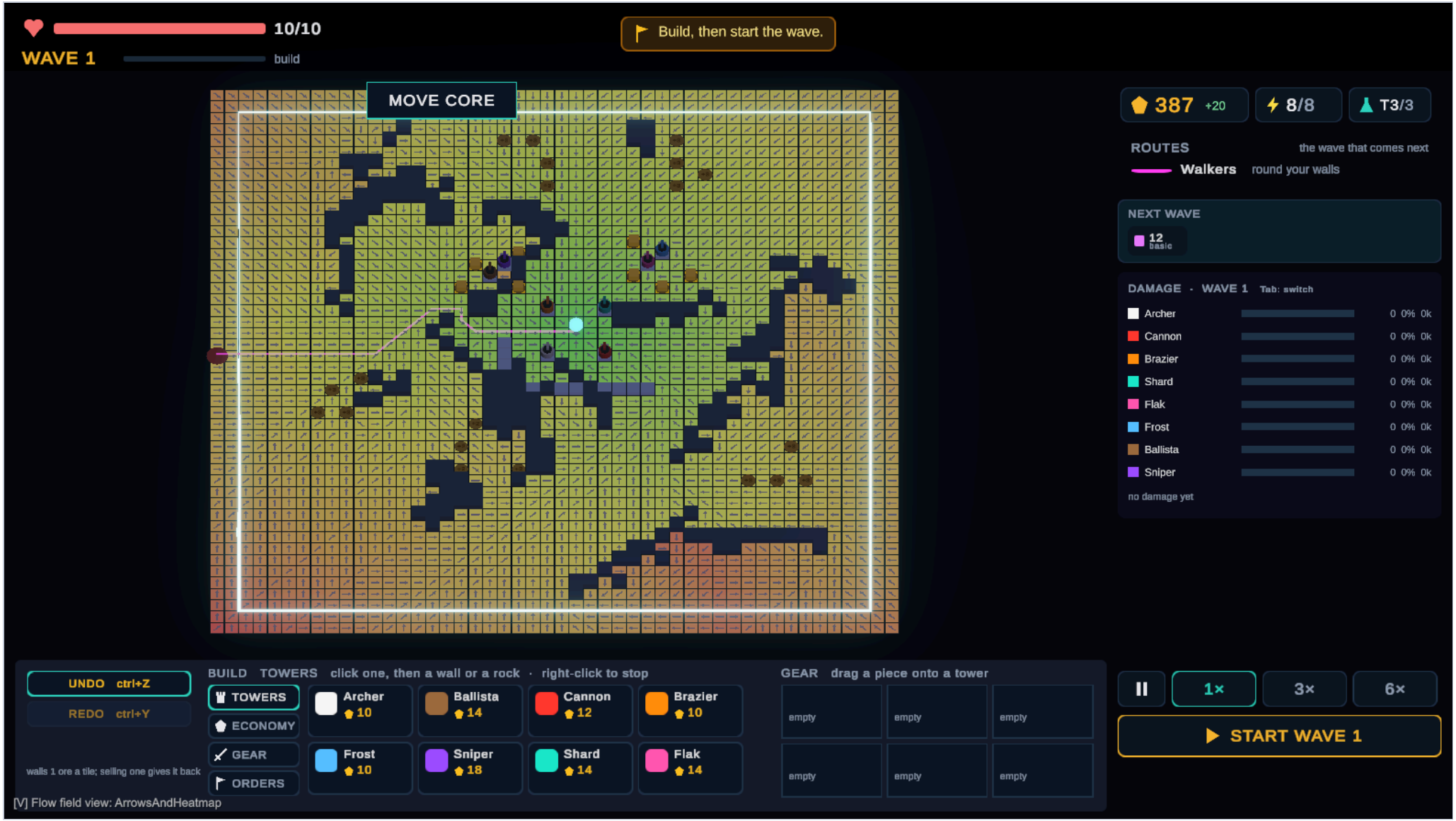
A wave in progress

FIGURE 6 **After a wave**



After a wave: the damage panel and the next wave from another side

FIGURE 7 **The flow-field view (V)**



The flow-field view

9.2 Still to come

Screen	Status
Session start: participant code	PLANNED (WP14)
Director overlay (F9): why the director chose what it chose; development only	PLANNED (WP14)
Preview of how the routes would change before a wall is paid for	PLANNED (second-semester UX pass)
Art: models for towers and enemies, sound	OPEN (question 13)

10. Controls

BUILT

Action	How
Build	pick it in the bottom bar, then click a tile; walls by click or drag
Build-menu tabs	click, or 1 to 4
Wall tool	Q
Collector	C
Sell	X, then click (drag over walls)
Move a tower	M, or drag the tower
Upgrade	U, or the inspect pane
Target order	T, or the inspect pane
Put the hand down	Esc, or the right mouse button
Undo, redo	Ctrl+Z, Ctrl+Y
Select a tower, collector, pylon or the core	click it
Start the wave	the wave button
Game speed	x1, x3, x6 buttons, or 1, 2, 3
Pause (during a wave only)	Space
Damage panel: this wave, the whole game, folded	Tab
Move the view, zoom	WASD or arrows, mouse wheel
Flow-field view	V, for development

A clash to fix: the keys 1 to 3 change the game speed *and* the build-menu tab at the same time (question 23).

11. Look and sound

Look. BUILT, as a placeholder: a dark board with a glowing edge; towers as figures with a colour each (white archer, brown ballista, red cannon, orange brazier, blue frost, violet sniper, teal shard, pink flak, lime pylon); walls coloured by tier; the interface drawn in code, with no art files. Enemies are coloured shapes.

Sound. OPEN (question 13). There is none.

12. Open questions

Each of these changes what the finished game is. "Suggestion" is this file's proposal, not a decision.

#	Question	Suggestion	Who decides	Needed by
1	**How long is a session, and what ends it?**	A fixed length of 25 waves; the session still ends early if the core falls; balance so that an ordinary player gets past wave 15. Reaching the last wave is the win.	you and your supervisor	the balance pass, firmly before the pilot
4	**Which wave types,** and what exactly is in each? (S10)	Pick four to six from the proposal's four and the development director's kinds (§7.3), one per weakness the profile can see.	you	WP9
5	**The exact formulas of the five measures.** (S9)	Start from `ARCHITECTURE.md` §5.3 and the proposal's Table 7.	you	WP8
6	**Raids and sieges change the player's income and firepower** (I11, S14).	Count raiders and siege enemies in the shared price list at the value of what they can destroy, or send the same raids in both versions.	you and your supervisor	WP9, before the pilot
7	**The banner in the static version.**	Both versions show one: the static one states the wave plainly ("Wave 7: armored and flyers"), the adaptive one also says why.	you and your supervisor	WP14
8	**How does a participant learn the game?**	A short practice run of three waves on a third seed, static, not analysed.	you and your supervisor	the pilot
12	**A name for the game,** and the company and product name in the build settings.	—	you	the first build that records data
13	**Art and sound.**	Distinct shapes for enemy types and a handful of sounds in the UX pass, if there is time.	you	the second-semester UX pass
14	**May a participant pause and change speed?**	Yes to both.	you and your supervisor	the pilot
20	**Do research, upgrades and gear stay?** A tester called research the worst feature and gear a waste of ore (2026-10-23).	Keep upgrades (they are the towers' identity); decide research and gear after the next tests.	you	the balance pass
21	**What does the static version send?** The fixed cycle of types (§7.1), or plain walkers?	The cycle: otherwise the two versions differ in variety, not in adaptation.	you and your supervisor	WP9
22	**Restarting waves in the study.**	Off in the study build, or counted and reported.	you and your supervisor	the pilot

#	Question	Suggestion	Who decides	Needed by
23	**Keys 1 to 3** change the speed and the build-menu tab together.	Give the tabs other keys or none (they lost their labels on 2026-10-28).	delegated	the next round

Answered

#	Question	Answer	When
2	Which towers?	Eight towers and the pylon (§5.2, §5.5).	2026-10-24
3	Which enemy types?	The eleven sent today (§6.3).	2026-10-14 to -26
9	Balance of chewed walls, refunds, rerolls	Chewed walls pay nothing; a sale gives half back, all of it in the build phase of the purchase; no shop, so no rerolls.	2026-10-20 to -28
10	Where do extra spawn points go?	All round the edge of the generated map; one to three a wave (§3.1).	2026-10-13
11	Is the "seeded maze" still wanted?	The map is generated from the seed (§3.1); the study's fixed seed sets keep both versions on the same maps.	2026-10-13
15	The walls	Single tiles by click and drag; three tiers since 2026-10-28.	2026-10-12
16	Rocks	Yes; generated, and breakable since 2026-10-28 (§3.3).	2026-10-12
17	One view for building and the wave?	Yes. The board is now seen at a tilt (§3.2).	2026-10-14
18	Which route lines are drawn?	One for each kind in the next wave in a build phase, the wave's kinds during it (§6.4).	2026-10-12
19	How big is the board?	48 × 48 (§3.1).	2026-10-20
—	Is the warning enough?	The wave is planned and shown as the build phase begins, so the player answers it before Start.	2026-10-12
—	Cards, levels and gear	Replaced by the build menu, upgrade paths and flat gear.	2026-10-20 to -27

13. From today to the finished game

Package	What the player will notice
Gameplay rounds (**ongoing**, to the end of October)	Whatever the next tester sessions call for.
WP7 the learner	Nothing. It runs without the game.
WP8 build profile	Nothing.
WP9 wave types and the price list	The final wave types; every wave of one number equally strong.
WP10, WP11 the director	Waves that answer the build, learned per player.
WP14 director on screen, session start	The participant-code screen; the director overlay for development.
WP12, WP13 harness and gates	Nothing.
Second semester: UX pass, balance pass	Route previews, a clearer screen, a game that is fair to an ordinary player.
Build freeze	Nothing changes after this, except crash fixes.

14. What this game is not

- **Not a game that gets easier or harder to suit you.** Difficulty is held equal on purpose; only the shape of the waves changes.
- **Not a game where walls block.** That was the original design and was removed; it is what makes "a path always exists" true.
- **Not a game of luck in what you can buy.** The random shop is gone; everything has a fixed price.
- **No generated text or behaviour.** Every banner line is written by hand, and every run can be replayed exactly.
- **Not in the plan:** anything carried over from one session to the next, more than one player.